RAKIBUL ISLAM RUMEL



Software Engineer – AI, Computer Vision, Full-Stack Systems

Rajshahi, Bangladesh Phone / WhatsApp: +880 1744-948674
Email: rumel9274@gmail.com Portfolio: rumel.vercel.app
GitHub: github.com/islamrakibul9274 LinkedIn: linkedin.com/in/rakibul-islam-817150248
ORCID: 0009-0007-1485-0144

PROFESSIONAL SUMMARY

Computer Science and Engineering (CSE) graduate from **CUET (CGPA: 3.43/4.00)** and published Computer Vision researcher (**IEEE COMPAS 2026**) combining solid CS theoretical foundations (DSA, OOP, DBMS, OS) with full-stack, 3D web, and production AI engineering. Experienced in shipping 10+ end-to-end applications across **FastAPI, YOLO, PyTorch, Next.js, React, Three.js, TypeScript, and Node.js**, alongside containerized backend infrastructure (**Docker, Redis, PostgreSQL, LangGraph**).

ACADEMIC CREDENTIALS & EDUCATION

Chittagong University of Engineering and Technology (CUET) — Chittagong, Bangladesh

B.Sc. in Computer Science and Engineering (CSE)
CGPA: 3.43 / 4.00 **Graduated: 2026**

- **Undergraduate Thesis:** *Fire Localization and Classification in Images Using Deep Learning* (Curated a custom object detection dataset with manually annotated bounding boxes for fire and smoke localization and classification).
- **Core CS Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, Artificial Intelligence & Machine Learning.

Higher Secondary Certificate (HSC) — Science **Passing Year: 2020**
Result: **GPA 5.00 / 5.00**

Secondary School Certificate (SSC) — Science **Passing Year: 2018**
Result: **GPA 5.00 / 5.00**

RESEARCH & PUBLICATIONS

Mish-Activated YOLOv8s for Wildfire and Smoke Detection (Paper ID: 427)

Conference: 2026 IEEE International Conference on Computing, Applications and Systems (**IEEE COMPAS 2026**)

Authors: Rakibul Islam Rumel, Maisha Fahmida
Presented: Dhaka University (October 9, 2026) ORCID: 0009-0007-1485-0144

- Published and presented peer-reviewed Computer Vision research in person at **IEEE COMPAS 2026**, receiving an official conference presentation certificate.
- Modified the YOLOv8s object detection architecture with **Mish** activation functions and **SPD-Conv** (Space-to-Depth Convolution) to preserve fine-grained small-object features, achieving **2.9 ms** preprocessing, **116.2 ms** inference, and **0.5 ms** postprocessing latency per image on an Apple Silicon M1 CPU.
- Curated a custom **2,843-image object detection dataset** (256×256 resolution, **70:15:15** train/validation/test split) by manually drawing bounding box annotations for wildfire and smoke detection, and subsequently extended the dataset for future instance segmentation research by converting the manual bounding boxes into high-precision polygon masks using box-prompted **SAM 2 Large (`sam2_l`)**.

TECHNICAL SKILLS

- **Computer Vision & Deep Learning: PyTorch**, **YOLO** (v5–v11), and **SAM 2** for object detection, instance segmentation, on-device Web Worker inference (ONNX Runtime Web), and model optimization (**SPD-Conv, Mish**).
- **Production AI Backends: Python**, **FastAPI**, and **LangGraph** for serving ML models, processing binary image streams, integrating high-speed LLM inference (**Groq**), and building stateful AI agents.
- **Full-Stack & 3D Web Development: Next.js**, **React**, **TypeScript**, **Three.js**, **React Three Fiber**, **Tailwind CSS**, and **Node.js / Express** for building end-to-end web applications.
- **Databases, DevOps & Core CS: PostgreSQL**, **MongoDB**, **Redis**, **Docker**, **Linux CLI**, and core Computer Science foundations (**DSA, OOP, DBMS, OS**).
- **AI Agentic Workflows: Claude Code**, **Codex**, **OpenCode**, **Gemini Antigravity** etc.

KEY ENGINEERING PROJECTS (RUMEL.VERCEL.APP)

Vision-Gate | Local AI Hazard Detection

Tech Stack: *Python 3.11 · FastAPI · YOLO · PyTorch* Project Overview — Local Inference Only

- Built a **FastAPI** backend to serve local **YOLO** models for fire and smoke detection, processing binary image streams and returning detection bounding boxes.

SwiftCare | ABA Care Team Workspace

Tech Stack: *Next.js · React · TypeScript · MongoDB · Auth.js · Groq · Recharts* Live Demo • GitHub Repository

- Developed clinic-scoped RBT, BCBA, and administrator workspaces with live session observations, server-saved drafts, treatment goals, and progress charts.
- Integrated optional AI-assisted SOAP/session drafts with clinician review and BCBA approval; added conflict-checked scheduling, access revocation, and an internal billing ledger with CSV export.

Facet | On-Device Computer Vision & 3D Interaction

Tech Stack: *Next.js · Three.js · React Three Fiber · ONNX Runtime Web · Zustand* Live Demo • GitHub Profile

- Created an interactive sculpture studio with four forms and four mineral finishes, adjustable animation, drag rotation, and responsive framing.
- Implemented optional YOLO person tracking in a local Web Worker, keeping camera frames on-device; lazy-loaded models and runtime assets with recoverable tracking errors.

NextFlight | Flight Discovery & Booking Platform

Tech Stack: *Next.js · React · TypeScript · MongoDB · Three.js · Stripe · Groq* Live Demo • GitHub Repository

- Built flight search with airport autocomplete, cabin options, flexible-date fare comparison, interactive seat selection, Stripe checkout, and booking management.
- Added flight inventory and booking administration, an AI travel concierge, and a scroll-driven Three.js fleet showcase with pause and reduced-motion support.

EcoSwap | Real-Time Plant Exchange Marketplace

Tech Stack: *Next.js · React · TypeScript · MongoDB · Pusher · JWT* Live Demo • GitHub Repository

- Developed searchable plant listings with image galleries, care details, availability filters, swap requests, wishlists, and personal inventory management.
- Integrated Pusher-powered messaging and a community leaderboard within a responsive interface with accessible controls and reduced-motion support.

INDUSTRIAL TRAINING & RESEARCH EXPERIENCE

Software & AI Engineering Trainee (Industrial Attachment) **November 2025**
Brain Machine Ai

- Completed a 2-week intensive undergraduate Software & AI Engineering Industrial Attachment at **Brain Machine Ai**, focusing on Python development, backend engineering workflows, and applied AI systems.

Undergraduate Computer Vision Researcher **2025 – 2026**
Department of Computer Science and Engineering (CSE), CUET

- Conducted deep learning research on wildfire and smoke localization, manually annotating 2,843 images for object detection and benchmarking custom Mish-activated YOLOv8s and SPD-Conv architectures.
- Presented Paper ID 427 in person at **IEEE COMPAS 2026**, Dhaka University (October 9, 2026).

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Education & Research Secretary **2024 – 2026**
Pabna Sirajganj Student Forum, Chittagong University of Engineering and Technology (CUET)

- Served as Member and elected Education & Research Secretary of the Pabna Sirajganj Student Forum at CUET.
- Managed academic workshop planning, educational resource dispatch, fresher orientation guidance, and student research coordination.

COMMUNICATION & PROFESSIONAL ATTRIBUTES

- **Languages:** Bengali (Native), English (Professional Working Proficiency in Technical Writing, Research Presentation, and Cross-Functional Collaboration).
- **Engineering Strengths:** End-to-end system ownership, clean API and database schema design, low-latency model optimization, and rapid full-stack delivery.

ACADEMIC REFERENCES

Maisha Fahmida

Lecturer, Department of Computer Science and Engineering (CSE)
Chittagong University of Engineering and Technology (CUET)
Chittagong-4349, Bangladesh
Email: maisha.fahmida@cuet.ac.bd
Relationship:Undergraduate Thesis Supervisor & Published Co-Author on IEEECOMPAS2026 Paper